# Complements to 10: Making Ten & Carry Trade

Addition Strategies to 10 • Standard: CCSS.MATH.1.OA.C.6

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**1. MATHEMATICAL CONCEPT & STACK APPROACH: Determine complements to 10 using Place Value Digit Blocks, Gold [+1] Carry Tokens, and RPN subtraction.**

Our base-10 system regroups every collection of 10 ones into 1 ten. Finding a number's complement to 10 (10 - 7 = 3) places the target base (10) on Shelf Y and the current ones digit (7) on Workbench X.

## 2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

**Published Block Kit: Place Value Studio (Digit Blocks & Gold [+1] Carry Tokens)**

Hands-On Setup: Place the [7] Digit Block into the Ones socket of the Place Value Tray and hold a Gold [+1] Carry Token beside the Tens socket.

**Student Prediction (Before Calculator): How many more ones must be added to [7] to reach 10 and trigger the Gold [+1] Carry Token? (7 + 3 = 10).**

## 3. CALCULATOR EXECUTION (SC-6 KEYCAPS & 4-LEVEL STACK TRACE)

**Keystroke Flow:** `1` `0` `ENTER` `7` `-`

| Key Pressed | Screen (X) | Internal 4-Level Stack Effect (T, Z, Y, X) |
|---|---|---|
| `1` | **1** | Enter 1 |
| `0` | **10** | Target ten (10) in X |
| `ENTER` | **10** | 10 lifted to Y |
| `7` | **7** | Current Ones digit (7) in X |
| `-` | **3** | 10 - 7 = 3: Adding 3 completes 10 and triggers the [+1] Carry! |

### 4. STORY CONTEXT (Making a Ten First (8 + 5 = 10 + 3 = 13) & The Gold [+1] Carry Token Trade)

Why is 10 + 3 so much easier for our brains than 8 + 5? Because base-10 place value groups everything into full tens! In Chapter 3 of From Blocks to the Stack, students bridge across 10 on the Place Value Tray and crown the Tens column with a Gold [+1] Carry Token.

**Core Investigation (Student Warm-Up A):**

The Ones socket holds [6]. How many more ones are needed to reach 10 and earn a Gold [+1] Carry Token?

# Bridging Through 10: The Make-Ten & Carry Strategy

Addition Within 20 • Standard: CCSS.MATH.1.OA.C.6

## 1. EXTENSION CONCEPT (WARM-UP B), BLOCKS & PREDICTION

**Concept & Stack Approach: Bridging across 10 decomposes an addend so the first part completes a full ten (8 + 5 = (8 + 2) + 3 = 10 + 3). On the stack, the intermediate sum (10) stays in X and automatically lifts to Y when the remaining part (3) is entered.**

Blocks & Prediction: Place [8] in Row 1 and [5] in Row 0 of the Ones column on the Place Value Tray. Prediction: Take 2 from the 5 to bring 8 up to 10. Place a Gold [+1] Carry Token in the Tens socket and [3] in the Ones socket: 10 + 3 = 13.

## 2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

8 ENTER 2 + 3 +

| Key Pressed | Screen (X) | Internal 4-Level Stack Effect |
|---|---|---|
| 8 | 8 | 8 in Ones socket |
| ENTER | 8 | 8 lifted to Y |
| 2 | 2 | 2 borrowed from 5 to make 10 |
| + | 10 | 8 + 2 = 10 (Gold [+1] Carry Token placed in Tens column!) |
| 3 | 3 | Remaining 3 ones |
| + | 13 | 10 + 3 = 13 total ([1] Ten, [3] Ones)! |

## 3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

**Misconception 1: Counting on by ones across 10 (e.g., 8 + 5 as 9, 10, 11, 12, 13) instead of decomposing 5 into 2 + 3 to complete 10 first.**

Hands-On Block Remediation: Have the student bridge [8] to 10 on the Place Value Tray, place a Gold [+1] Carry Token in the Tens socket with [3] in the Ones socket, and key [8] [ENTER] [2] [+] [3] [+].

**Misconception 2: Reading teen numbers as two separate single digits (seeing 14 as 1 and 4 rather than 1 Ten plus 4 Ones).**

Hands-On Block Remediation: Place [1] in the Tens socket and [4] in the Ones socket of the Place Value Tray and key [10] [ENTER] [4] [+] so students see 14 formed from 10 in Register Y and 4 in Register X.

## 4. EXTENSION MATHEMATICAL INSIGHT

Bridging across 10 decomposes an addend so the first part completes a full ten (8 + 5 = (8 + 2) + 3 = 10 + 3). On the stack, the intermediate sum (10) stays in X and automatically lifts to Y when the remaining part (3) is entered.

# Complements to 10 & Make-10 Regrouping with Carry Tokens

Name: _____ Date: _____ Class: _____

### STUDENT REFERENCE GUIDE: COMPLEMENTS TO 10 & GOLD [+1] CARRY RULES

- The Ones column holds single Digit Blocks [0] through [9]. The moment a column reaches 10, it regroups!
- 10-for-1 Carry Trade: When the Ones column reaches 10, place a Gold [+1] Carry Token into the Tens column and reset the Ones socket to [0].
- Complements to 10: To find how much more is needed to trigger a Gold [+1] carry, enter 10 [ENTER], then the digit, then [-].

**Key Sequence:** `1` `0` `ENTER` `7` `-`

**Investigation 1: What is the complement to 10 for 4? Write the keystrokes.**

**1. Draw Your Blocks on the Tray:**

**2. Number Bond & Two Shelves:**

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Checked on SC-6**

**Investigation 2: Solve 7 + 5 using Make-Ten: first make 10, then add 2.**

**1. Draw Your Blocks on the Tray:**

**2. Number Bond & Two Shelves:**

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Checked on SC-6**

**Investigation 3: The Ones socket holds [9]. How many more to trigger a Gold [+1] Carry Token?**

**1. Draw Your Blocks on the Tray:**

**2. Number Bond & Two Shelves:**

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Checked on SC-6**

**Investigation 4: Solve 8 + 6 using Make-Ten on the calculator.**

**1. Draw Your Blocks on the Tray:**

**2. Number Bond & Two Shelves:**

Top Shelf (Line Y):

Bottom Shelf (Line X):

Combine (+ or −):

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Checked on SC-6**

# Unit 03: Complements to 10 & Make-10 Regrouping with Carry Tokens — Answer Key

Complete solutions and SC-6 verification keystrokes for Grade 1

## 1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

**Core Investigation (Warm-Up A): The Ones socket holds [6]. How many more ones are needed to reach 10 and earn a Gold [+1] Carry Token?**

Core Solution: 4 ones. Keystrokes: [1][0] [ENTER] [6] [-]

**Core Check Keys:** 1 0 ENTER 6 -

**Extension Challenge (Warm-Up B): Use the Make-Ten strategy to solve 9 + 6 on your Place Value Tray and calculator.**

Extension Solution: 15. Keystrokes: [9] [ENTER] [1] [+] [5] [+]

**Ext. Check Keys:** 9 ENTER 1 + 5 +

## 2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

**Investigation 1: What is the complement to 10 for 4? Write the keystrokes.**

**Expected Answer: 6**

**SC-6 Verification Keys:** 1 0 ENTER 4 -

**Investigation 2: Solve 7 + 5 using Make-Ten: first make 10, then add 2.**

**Expected Answer: 12**

**SC-6 Verification Keys:** 7 ENTER 3 + 2 +

**Investigation 3: The Ones socket holds [9]. How many more to trigger a Gold [+1] Carry Token?**

**Expected Answer: 1**

**SC-6 Verification Keys:** 1 0 ENTER 9 -

**Investigation 4: Solve 8 + 6 using Make-Ten on the calculator.**

**Expected Answer: 14**

**SC-6 Verification Keys:** 8 ENTER 2 + 4 +

### Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.